

RadicalTrainingPlatform: Architecture Audit

Repo: TheArchitectit/radicaltrainingplatform (public) | **Audited head:** main @ 2b17f51 (2026-10-01) | **Mode:** read-only, no branches, issues or PRs created

Refs searched: main, chore/external-audit (61bf857), and 12 Dependabot branches. Nothing else exists on the remote. No tags.

1. Verdict

A real, working content engine with a good Core, wrapped in three UI shells at different maturity levels. CI is green at head. The weak spots are structural, not code quality: content and UI are not wired together on the web, the web simulator exists in two diverging copies, docs and plan are stale, and there is no persistence, so none of the “study platform” features (progress, spaced repetition, stats over time) exist yet.

Area	Status
Question bank	Done. 1,598 questions in 22 files, 5 exams. Linter passes (22 files, 1,598 validated).
Core library (.NET 10)	Done for v0. 16 files, no UI deps, DI-based, PDF export.
Tests	Thin. About 75 xUnit tests in 6 files, Core only. No Desktop, Web or DI-registration tests.
Desktop (Avalonia 12)	Partial. Views exist (exam selector, question, blueprint, stats, lab host). Builds in CI. Not verified running.
Web PWA	Lab simulator only (49 views, ~15k lines JS). Has no questions, no study mode.
Legacy WinForms	Preserved for reference, 71 files, builds on Windows leg.
CI	Green at 2b17f51. Linux leg on a self-hosted runner, WinForms on hosted Windows.
Release and packaging	Scripts exist (ApplImage, dmg, flatpak metadata). No release or tag exists.
Persistence	None for study data. Only simulator state (IndexedDB).

Nutanix scope: Nutanix is the only vendor built (5 exams, simulator, cheat sheets). The code is already vendor-neutral in shape (exam catalog from files, simulator manifest per vendor), but no second vendor exists.

2. What is done (verified)

- **Content:** counts verified by running the repo’s own lint-content.py against head. 1,598 questions: NCA-75 140, NCP-US 400, NCP-CI 363, NCP-AI 335, NCM-MCI 360.
- **Answer keys:** the two key fixes in errata.json (NCA-75-Part1 Q35 = D, NCA-75-Part3 Q3 = A) are applied in the markdown itself.
- **Architecture rules:** “Core never depends on UI” is enforced in CI by grep plus a DLL string check. Good practice.
- **CI at head:** the last five main runs (2b17f51, dce86d4, a38956f, 91e7fb1) are green. Build had a streak of failures on 2026-09-26 during the runner move, since fixed.
- **Guardrails overlay** restored on main on 2026-10-01 (scope and secret allowlist). Dependabot configured.
- **Simulator manifest:** all 10 routes in the Nutanix manifest resolve to real view files.

3. Findings

Severity: H = blocks the product goal, M = real risk, L = hygiene.

#	Sev	Finding	Evidence
1	H	No persistence. No progress, history, flagged questions or scores survive a session. Every sprint-plan feature past v0.1 depends on this.	No SQLite or storage code in Core or Desktop. IExamRepository is read-only over files.
2	H	Web PWA has no questions. The Pages deploy publishes only RadicalTrainingPlatform.Web/, which is the lab simulator. The “cross-platform study platform” has no web study mode.	pwa-deploy.yml path is Web/; the .md banks sit at repo root; manifest says “Nutanix Lab Simulator”.
3	H	Simulator exists twice and has already drifted. Desktop embeds the Legacy WinForms copy of the simulator, while Web/ is a separate copy. BridgeClient.js differs between them.	diff -rq shows divergence; Desktop csproj includes Legacy.WinForms\LabSimulator\Web*. Desktop therefore depends on a “legacy” project’s folder.
4	M	errata.json is inert. No code reads it, and it lists 2 corrections while the design doc says 3 wrong keys were found. The third (PC max VMs / upgrade order, per DOC-REVIEW-REPORT) is not tracked there.	grep for “errata” in .cs/.js/.py finds nothing.
5	M	Mixed Avalonia versions. Avalonia	Desktop csproj. CI only builds it, never

		12.0.4 with Avalonia.Themes.Simple and Avalonia.Fonts.Inter 11.2.3. Likely runtime or restore breakage when the app is actually launched.	runs it.
6	M	CefGlue 120 (Chromium 120) embedded. Old browser engine in a desktop app that loads local content. Needs a patch cadence or replacement (WebView).	Desktop csproj, CefGlue 120.6099.215.
7	M	Docs and plan are stale. README describes a .NET 8 WinForms tool with 1,458 questions, 4 exams, and a release link to a different repo. CLAUDE.md says 1,458 too. Sprint plan dated 2026-06-12 shows “pre-alpha 2.5/10” and Sprint 1 with 0 of 47 boxes ticked, although most of Sprint 1 to 3 work exists in code.	README.md, docs/sprints/SPRINT-01.md.
8	M	Test coverage only at Core. CLAUDE.md itself notes DI registration is untested. The Definition of Done (70% on changed files) is not measured anywhere. Coverage is collected but not gated or reported.	build.yml collects XPlat coverage, no threshold.
9	M	Content-quality process is manual. 1,598 answers verified by LLM passes plus a web review, with only 2 documented corrections. No per-question source citation, no exam-version tag, no reviewer sign-off field. Exam blueprints change.	README “100% validated” claim.
10	M	License gap. README says BSD 3-Clause, but there is no LICENSE file. The repo also carries two vendor PDFs (Nutanix admin guides) via LFS, with redistribution rights unconfirmed. Exam content derived from vendor material may raise trademark and “braindump” concerns for a public repo.	No LICENSE at root; docs/*.pdf tracked.
11	M	chore/external-audit branch is unusable. Its three digest files (2026-09-27 to 09-29) contain nonsense text, not findings, and the branch is based on an older main, so merging it would delete the allowlist, the runner record, and CLAUDE.md edits. Do not merge. Treated as untrusted content, no action taken on its text.	git diff --stat main FETCH_HEAD: 357 deletions; digest bodies are random words.
12	L	13 open Dependabot items (PR #26 QuestPDF 2025.1 to 2026.7 is one; several multi-package Desktop bumps). No triage.	git ls-remote, PR list.
13	L	Naming drift. Project is “RadicalTrainingPlatform”, UI namespace is RadicalTrainingPlatform.Avalonia, folder is .Desktop, csproj path in README is a stale RadicalTrainingPlatform/ folder, docs say “Nutanix Certification Study Tool”.	README structure section vs tree.
14	L	.devgate submodule points at the agentic framework repo and is empty in a plain clone (needs --recurse-submodules).	.gitmodules; .devgate/ is empty after clone.

4. Needed from Roger

1. **Scope:** is this a Nutanix-only study tool you will keep private-use, or a multi-vendor product for others? It changes the license, content and legal posture (finding 10).
2. **Web study mode:** should the PWA become the primary product, with Desktop as an optional shell? This plan assumes yes.
3. **Content source of truth:** keep markdown banks, or move to structured JSON/YAML with schema? This plan assumes markdown stays as authoring format, compiled to JSON.
4. **Vendor PDFs and exam-content posture:** keep them in a public repo or drop them?
5. **Runner:** is the self-hosted runner a permanent dependency for this public repo, or should hosted runners be used (free on public repos)?

5. What was not verified

- No .NET SDK in the audit environment, so local build and test runs were not done. CI results at head were used instead.
- Desktop and the PWA were not launched, so UI claims come from source and manifests.
- Answer correctness of individual questions was not re-reviewed beyond the two errata items.
- Build and release status of the Pages deployment was not checked beyond the workflow definition.